



PhD Position in Wearable Robotics & Control (WeRoCon) Laboratory

Department of Electrical Engineering, Indian Institute of Technology Jodhpur

Overview

The Wearable Robotics & Control (WeRoCon) Laboratory at the IIT Jodhpur invites applications for a fully funded PhD position in developing intelligent and robust control strategies for lower-limb prostheses and exoskeletons. The successful candidate will work on the design and development of these wearable robots, with a focus on implementing and experimentally validating advanced control strategies for powered prosthetic legs and exoskeletons.

The project will involve close collaboration with the best clinical partners in the country at All India Institute of Medical Sciences, Jodhpur (AIIMS) and direct interaction with individuals with lower-limb loss and mobility impairments, including those arising from spinal cord injury, post-stroke hemiparesis, multiple sclerosis and other neuromuscular or orthopedic conditions affecting gait. The candidate will be responsible for deploying rehabilitation robots in real-world scenarios and systematically evaluating their impact on users' mobility, safety, comfort and overall experience.

Applicant Profile

The ideal candidate should have a strong academic background in mechatronics and control systems, along with hands-on experience with robot hardware and integrating control algorithms.

Required Qualifications:

- GATE qualified (Students from NIT are exempted)
- Proficiency in MATLAB, Simulink, Python, C++, Linux, and ROS
- Solid foundation in robotics, control theory with strong analytical skills
- Excellent communication abilities and demonstrated teamwork in interdisciplinary settings

Desirable Qualifications:

- Experience with motion capture, human-in-loop testing, and lower-limb wearable robots
- Background in machine learning/ optimization for real-time control and statistical data analysis
- Peer-reviewed publications in robotics conference

Application Instructions

Interested students are requested to contact me with a cover letter and their most recent CV. Review of applications begins immediately and continues until the position is filled.

For more information regarding the position, please reach out to me directly.

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